

Universal Quadratic Image Containment over F_2

A Computer-Assisted Proof that $h(6) \geq 25$

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Public reproducibility repository: <https://github.com/RicardoVera-Lab/h6-universal-quadratic-image-theorem>

Status note

This manuscript reports an internally hardened computer-assisted mathematical result. The proof/certificate chain has passed a clean-room reconstruction and a second internal checker, but it has not yet undergone independent third-party mathematical reproduction or journal peer review. The main claim is $h(6) \geq 25$; no claim of $h(6)=25$, $h(6) \geq 26$, or absolute historical priority is made.

Abstract

Let $h(n)$ be the least cardinality of a set $H \subset F_2^{(2n)}$ that is not contained in the image of any map $q: F_2^n \rightarrow F_2^{(2n)}$ whose coordinate functions have algebraic degree at most two. We prove, by a combination of human-readable reductions and exact finite certificates, that $h(6) \geq 25$. Equivalently, every set of at most 24 points in F_2^{12} is contained in the image of some coordinatewise quadratic map $F_2^6 \rightarrow F_2^{12}$. The key reformulation factors every such map through the 22-dimensional degree- ≤ 2 evaluation embedding $\phi_2: F_2^6 \rightarrow F_2^{22}$ and turns the problem into adaptive matching of right-kernel relations. For 24 targets, two evaluation dependencies are unavoidable, whereas the target matrix has a kernel of dimension at least 12. The proof shows that a realizable 2-dimensional dependency space can always be placed inside that target kernel. The finite part consists of 46 dangerous two-dimensional weight profiles, 45 exact-kernel witness pairs, nine exact rational Delsarte/Krawtchouk certificates, and an exhaustive MacWilliams feasibility closure, with a separate all-one exceptional branch. All proof-authoritative computation uses exact integer or rational arithmetic. A public verifier and witness corpus are provided for reproduction.

Keywords: Reed–Muller codes; Boolean functions; finite geometry; binary linear codes; computer-assisted proof; Delsarte bounds; MacWilliams identities; sum-free sets; interpolation.

1. Introduction

Quadratic Boolean maps occupy a natural boundary between linear structure and nonlinear expressivity. In six Boolean variables, the vector space of scalar functions of algebraic degree at most two has dimension $1+6+C(6,2)=22$. If one fixes the preimage locations in

advance, this immediately suggests a 22-point interpolation barrier: beyond 22 selected domain points, linear dependencies among the evaluation columns are unavoidable. The problem considered here is different. The preimages are allowed to depend on the target set. That adaptivity changes the geometry of interpolation. Instead of requiring an evaluation matrix with trivial kernel, it is enough to choose evaluation points whose unavoidable relations are relations already satisfied by the target vectors. The distinction is small algebraically but decisive combinatorially.

This paper establishes a two-point excess beyond the naive 22-dimensional barrier. For every set H of at most 24 points in F_2^{12} , one can choose 24 preimages in F_2^6 and a linear projection of the quadratic evaluation configuration so that all points of H appear in the image. We refer to the conceptual mechanism as adaptive dependency matching: the proof does not create new polynomial degrees of freedom, but places the forced evaluation dependencies inside the target dependency space.

The argument combines four layers. First, a direct Reed–Muller/evaluation equivalence converts image containment into a right-kernel containment problem. Second, a hypothetical bad target yields an 11-dimensional even binary code of minimum distance at least four. Third, nearly all dangerous 2-dimensional dependency profiles are shown by exact certificates to occur as exact kernels of 24 selected quadratic evaluation points. Finally, if the target code avoided all such realizable 2-planes, its dangerous words would form a very large sum-free set; a structural theorem for large sum-free subsets of binary vector spaces then forces a dimension-10 subcode with only safe weights, which exact MacWilliams constraints rule out.

The computational component is deliberately certificate-based. The verifier checks exact ranks, exact witness orthogonality, exact rational Delsarte inequalities, and exact integral MacWilliams feasibility. No floating-point comparison, random success, timeout, or solver absence is used as a proof of impossibility.

2. Definition and main theorem

Definition 2.1. For $n \geq 1$, define

$h(n) = \min\{|H| : H \subset F_2^{2n} \text{ and, for every coordinatewise algebraic-degree-}\leq 2 \text{ map } q: F_2^n \rightarrow F_2^{2n}, H \text{ is not contained in } \text{im}(q)\}.$

Here each coordinate of q is a Boolean polynomial in algebraic normal form of degree at most two.

Theorem 2.2 (main theorem). Every set $H \subset F_2^{12}$ with $|H| \leq 24$ is contained in the image of some coordinatewise quadratic map $q: F_2^6 \rightarrow F_2^{12}$. Consequently,

$h(6) \geq 25$.

The theorem is a lower bound only. It does not determine $h(6)$ exactly.

3. The quadratic evaluation embedding

Let P_2 be the 22-dimensional F_2 -vector space of Boolean polynomials in six variables of degree at most two. Fix the monomial basis

$$M_2 = \{1, x_1, \dots, x_6, x_i x_j : 1 \leq i < j \leq 6\}.$$

Define the evaluation embedding $\varphi_2: F_2^6 \rightarrow F_2^{22}$ by $\varphi_2(x) = (m(x))_{m \in M_2}$. Its first coordinate is 1 and the next six linear coordinates recover x , so φ_2 is injective. The 64 evaluation columns span F_2^{22} : a coefficient vector orthogonal to all 64 columns would define a degree- ≤ 2 Boolean polynomial vanishing on all of F_2^6 , hence the zero polynomial by uniqueness of algebraic normal form.

Lemma 3.1 (factorization).

Every coordinatewise degree- ≤ 2 map $q: F_2^6 \rightarrow F_2^{12}$ factors uniquely at the coefficient level as $q(x) = T\varphi_2(x)$ for some linear map $T: F_2^{22} \rightarrow F_2^{12}$. Conversely, every linear T defines such a quadratic map.

For an ordered set $X = (x_1, \dots, x_t)$ of distinct domain points, let $V_X = [\varphi_2(x_1) \dots \varphi_2(x_t)]$ be the $22 \times t$ evaluation matrix. For an ordered target list $Y = (y_1, \dots, y_t)$, let $M_Y = [y_1 \dots y_t]$ be the $12 \times t$ target matrix.

Lemma 3.2 (kernel criterion).

There exists a linear map T with $M_Y = TV_X$ if and only if $\ker(V_X) \subset \ker(M_Y)$.

Proof. The equation $M_Y = TV_X$ is solvable exactly when every row of M_Y lies in the row space of V_X . Over F_2 , the right kernel of a matrix is the orthogonal complement of its row space. Hence $\text{row}(M_Y) \subset \text{row}(V_X)$ if and only if $\ker(V_X) \subset \ker(M_Y)$. ■

This gives an exact equivalence: a target set is contained in the image of a quadratic map if and only if one can choose distinct evaluation points whose dependency space is contained in the target dependency space.

4. Why the barrier at 22 can be exceeded

For $t \leq 22$, choose t independent evaluation columns. Then $\ker(V_X) = 0$, and the kernel criterion is automatic for every target matrix. Thus the theorem is trivial through 22 points.

For 23 or 24 selected evaluation points, nontrivial relations are unavoidable. At 24 points, a full-rank evaluation matrix has rank 22 and therefore a 2-dimensional right kernel. The target matrix is much more redundant: a 12×24 matrix has kernel dimension at least 12. The proof exploits this surplus.

Conceptually, the problem is: given the large target kernel $K = \ker(M)$, can one find a 2-dimensional subspace $L \subset K$ that is the exact right kernel of 24 columns from the quadratic evaluation configuration? The proof shows that one can.

This is the two-point excess mechanism:

22 scalar quadratic degrees of freedom + 2 unavoidable evaluation relations, adaptively matched to ≥ 12 target relations.

5. Reduction from a bad target to an even code

It is enough to prove the result for exactly 24 targets: any smaller set can be enlarged to a 24-set in F_2^{12} .

Assume for contradiction that a 24-set H is not contained in the image of any coordinatewise quadratic map. Translate H by a vector outside H . Output translation preserves degree and non-interpolability, so we may assume that the 24 targets are distinct and nonzero.

Order them as $h_1, \dots, h_{24}$ and form $M = [h_1 \dots h_{24}] \in F_2^{12 \times 24}$. Let $K = \ker(M)$. Then $\dim K \geq 12$. Because all target columns are nonzero and distinct, K has no word of weight one or two; hence $d(K) \geq 3$.

Intersect K with the even-weight hyperplane. Since that hyperplane has codimension one, K contains an 11-dimensional even subspace E . Every nonzero word of E has weight at least four. Delete coordinates that are identically zero on E . The resulting code, still denoted E , is a full-support even $[N, 11, \geq 4]$ code with the same weights.

Puncturing one coordinate is injective because $d(E) \geq 4$, giving an $[N-1, 11, \geq 3]$ code. The radius-one Hamming bound gives $2^{11} \cdot N \leq 2^{N-1}$. This fails at $N=15$ and is an equality at $N=16$. Therefore $16 \leq N \leq 24$.

6. Dangerous weights and two-dimensional profiles

Define the dangerous set of E by

$$D(E) = \{c \in E \setminus \{0\} : \text{wt}(c) \in \{8, 12, 14, 16, 18, 20, 22, 24\}\}.$$

The remaining nonzero even weights compatible with the preceding constraints are $\{4, 6, 10\}$; we call these safe weights.

A two-dimensional binary code has three nonzero words. If their weights are (a, b, c) , then, up to relabeling, the three nonzero coordinate-type multiplicities are $(a+b-c)/2$, $(a+c-b)/2$, and $(b+c-a)/2$, and the union support has size $u = (a+b+c)/2$. Thus a sorted weight triple determines the two-dimensional code up to coordinate permutation.

Exhaustive exact enumeration of sorted triples drawn from the dangerous weights, subject to nonnegative integral coordinate multiplicities and $u \leq 24$, yields exactly 46 possible profiles.

7. Exact-kernel certificates for 45 profiles

The public witness file contains 45 exact-kernel witness pairs. Each record specifies a dangerous profile and two 64-bit truth tables f and g .

For each record, the verifier checks all of the following exactly: the weights of f , g , and $f+g$ equal the advertised profile; each dependency is orthogonal to every degree- ≤ 2 monomial evaluation row; the union support U has the expected size; the evaluation matrix restricted to U has rank $|U|-2$; hence its right kernel is exactly $\text{span}(f|_U, g|_U)$; and the support can be extended by rank-increasing evaluation columns to 24 distinct points of total rank 22 without changing those dependencies.

The unique dangerous profile without such a witness is $(12, 12, 24)$. On 24 coordinates, the weight-24 word is the all-one vector, so this profile is isolated as the exceptional all-one geometry.

A witness for a sorted profile applies to any actual 2-plane with the same nonzero weights: the weight triple fixes the three coordinate-type multiplicities up to permutation and basis change in $GL(2,2)$. Reordering the 24 selected points transfers the exact-kernel realization to the actual 2-dimensional subcode.

8. From a bad target to a large sum-free set

Suppose $c, d, c+d$ are three dangerous nonzero words in E and their profile is not $(12, 12, 24)$. Then the exact-kernel certificate provides 24 distinct evaluation points X with $\ker(V_X) = L = \text{span}(c, d)$. Since $L \leq E \leq K = \ker(M)$, the kernel criterion gives $M = TV_X$ for some T , contradicting the assumption that H is bad.

Therefore, if the all-one vector is not in E , $D(E)$ must be sum-free. If the all-one vector belongs to E , then $D(E) \setminus \{1\}$ must be sum-free: removing 1 eliminates the only exceptional dangerous profile.

9. Exact Delsarte lower bounds

Consider first the generic branch $1 \notin E$. Let A_i be the weight distribution of the full-support even $[N, 11, \geq 4]$ code E . Then $A_0 = 1$; odd A_i vanish; $A_2 = 0$; the nonzero weight counts sum to 2047; and full support implies $B_1 = 0$ in the dual weight distribution. MacWilliams/Delsarte nonnegativity gives the usual Krawtchouk inequalities for $B_j \geq 0$.

For each active length $N \in \{16, \dots, 24\}$, an exact rational dual certificate supplies α, β and $\lambda_j \geq 0$ such that

$$\alpha + \beta K_1(i) + \sum_j \lambda_j K_j(i) \leq 1_D(i)$$

for every allowed even weight i , where 1_D is the indicator of the dangerous weight set.

Multiplying by A_i and using the MacWilliams constraints yields an exact lower bound for $|D(E)|$.

The certified bounds are:

$N=16$: 1011

$N=17$: 938

$N=18$: 910

$N=19$: 9532/11

$N=20$: 746

$N=21$: 4033/5

$N=22$: 48555/56

$N=23$: 177445/212

$N=24$: 1680251/1865

Every bound is strictly greater than 640. All inequalities are checked with exact rational arithmetic.

10. Large sum-free structure

We use the binary case of a theorem stated in Leo Versteegen's 2024 work on large sum-free sets: if $A \subset \mathbb{F}_2^n$ is sum-free and $|A| > 5 \cdot 2^{n-4}$, then A is contained in a coset of a proper subspace. For $n=11$ the threshold is 640.

Identify the 11-dimensional code E with F_2^{11} . In the generic branch, $D(E)$ is sum-free and has more than 640 elements, so $D(E) \subset x+V$ for some proper subspace $V < E$. The coset cannot be V itself: a sum-free subset of a vector space V has size at most $|V|/2$, while a proper subspace of an 11-dimensional space has at most 1024 points, giving at most 512 sum-free points.

Thus $x \notin V$. Choose a nonzero linear functional $\ell: E \rightarrow F_2$ with $\ell(V)=0$ and $\ell(x)=1$. Then $C=\ker(\ell)$ is a dimension-10 hyperplane disjoint from the dangerous coset. Every nonzero word of C therefore has safe weight in $\{4,6,10\}$.

11. MacWilliams closure of the safe-weight branch

Delete any coordinates identically zero on C . The resulting code remains dimension 10, full support, and has nonzero weights only in $\{4,6,10\}$. A Hamming-bound argument gives active length at least 15.

Let A_4, A_6, A_{10} be the nonzero weight counts. They satisfy $A_4 + A_6 + A_{10} = 1023$ and the full-support condition $B_1 = 0$. For every nonnegative integer candidate satisfying these basic equations, the verifier computes every dual MacWilliams coefficient

$$B_j = 2^{-(10)} \sum_i A_i K_j(i)$$

and requires it to be a nonnegative integer. Exhaustion over the relevant lengths leaves zero feasible candidates. Therefore no full-support dimension-10 binary code with all nonzero weights in $\{4,6,10\}$ exists. This contradicts the hyperplane forced by the large sum-free theorem.

12. Exceptional all-one branch

Now suppose E has length 24 and contains the all-one word 1 . The involution $c \mapsto c+1$ partitions the 2048 words of E into 1024 complement pairs. Allowed even weights pair as $0 \leftrightarrow 24$, $4 \leftrightarrow 20$, $6 \leftrightarrow 18$, $8 \leftrightarrow 16$, $10 \leftrightarrow 14$, and $12 \leftrightarrow 12$. Every complement pair contains at least one dangerous word. Hence $|D(E)| \geq 1024$ and $|D(E) \setminus \{1\}| \geq 1023 > 640$.

By the exact-kernel argument, $D(E) \setminus \{1\}$ is sum-free. Apply the same large sum-free theorem and separating-functional argument to obtain a dimension-10 hyperplane C disjoint from $D(E) \setminus \{1\}$.

If $1 \notin C$, all nonzero weights of C lie in $\{4,6,10\}$, contradicting the safe-weight MacWilliams closure. If $1 \in C$, then C is full support of length 24 and its allowed nonzero weights are $\{4,6,10,24\}$, with the unique weight-24 word appearing once. In the B_1 numerator, the zero word and all-one word cancel because $K_1(0)=24$ and $K_1(24)=-24$, while $K_1(4)=16$, $K_1(6)=12$ and $K_1(10)=4$ are all positive. The remaining 1022 nonzero words therefore force $B_1 > 0$, contradicting full support.

13. Proof of the main theorem

The generic branch and the all-one branch both contradict the existence of a bad 24-point target set. Therefore every 24-set in F_2^{12} is contained in the image of some coordinatewise degree- ≤ 2 map $F_2^6 \rightarrow F_2^{12}$. Every smaller set can be enlarged to a 24-set, so the conclusion holds for all $|H| \leq 24$. Hence $h(6) \geq 25$. ■

14. Computational certificate architecture

The proof package is designed so that the finite claims can be checked without trusting the code that originally generated them.

Primary verifier: `certificate/verify_h6_lower25_cleanroom.py`

Second internal checker: `certificate/verify_h6_publication_independent.py`

Witness corpus: `certificate/h6_exact_profile_witnesses.txt`

The primary verifier checks the rank-22 evaluation configuration, regenerates the 46 dangerous profiles, validates all 45 exact-kernel witness pairs, verifies the nine exact rational Delsarte certificates, exhausts the safe-weight MacWilliams candidates, closes the all-one branch, and runs negative self-tests against corrupted data.

The second checker does not import the primary code. It changes several local mathematical implementations: evaluation columns rather than row bitsets; fourth finite differences rather than Möbius/ANF transformation for degree tests; column-pivot rank rather than row elimination; and Krawtchouk coefficients extracted from $(1+z)^{(n-i)}(1-z)^i$ rather than the primary binomial-sum implementation. The witness corpus and Delsarte coefficient data are nevertheless shared, so this is internal reimplementations rather than external independence.

Frozen SHA-256 values:

Primary clean-room verifier:

`03b080ea2b24fda30e27a08b5fd1c52d859cb4a113342ddaa84ce3803557cc9f`

Witness corpus:

`75151f69a7325673c3270fa0aab35a222bb0263f6ab7ae26b953bf45d10c5eac`

Expected terminal verdicts are `INTERNAL_COMPUTATIONAL_VERDICT=PASS` and `INDEPENDENT_CHECK_VERDICT=PASS`.

15. Adversarial validation

The publication-hardening audit explicitly attacked the proof for off-by-one errors, repeated or zero targets, translation normalization, incorrect kernel dimension, minimum-distance failure, even-subcode dimension, active-length bounds, incomplete profile enumeration, contained-kernel versus exact-kernel confusion, circular basis extension, an unhandled all-one profile, floating-point or sign errors in the Delsarte step, misuse of the strict 640 threshold, an invalid coset-to-hyperplane inference, incomplete MacWilliams feasibility, and hidden use of the desired conclusion. No attack produced a counterexample or unresolved bridge within the frozen scope.

Negative self-tests also reject a one-bit-corrupted witness, dependent duplicate generators, the impossible near-profile (8,8,24), an invalid inflated Delsarte certificate, and a fake weight enumerator satisfying only superficial constraints.

16. Residual limitations

The mathematical status of the frozen proof is `CONFIRMED_WITHIN_SCOPE`, but several residual boundaries remain.

First, the large sum-free theorem is taken from the literature rather than reproved or formally verified here. Second, both internal verifiers use the same 45-witness corpus and the same

Delsarte coefficient data. Third, the proof is not formalized in Lean, Coq, Isabelle, or another proof assistant. Fourth, no competent external third party has yet independently reconstructed the theorem. These limitations affect the maturity and independence of the result, not an identified internal mathematical gap in the present certificate chain.

17. Prior-art and originality note

A targeted prior-art search through formulations involving quadratic maps, vectorial Boolean functions, Reed–Muller evaluation/interpolation, finite geometry, and polynomial images did not locate a published statement equivalent to the universal ≤ 24 containment theorem proved here. This observation should be read narrowly: failure to locate an equivalent result in a targeted search is not a proof of historical priority.

Separate structural objects investigated during the project have different prior-art states and are intentionally not claimed as contributions of this paper. In particular, the isolated binary $[14,3,8]$ duplicated-simplex object is classical, and the ambient universe of projective binary $[16,7]$ codes has been enumerated previously. A finer 71-class forbidden-subconfiguration census and a related 17-point threshold remain separate research objects with distinct novelty and proof-status boundaries.

18. Discussion: adaptive dependency matching

The result is best understood as a statement about where dependencies can be placed. The 22-dimensional function space does not change. What changes is the order of quantifiers: the target set is given first, and the 24 preimages are chosen afterward. This allows the two relations forced by rank 22 to be aligned with relations that already exist in the target kernel. The proof reveals a broader research question. For general n , the scalar quadratic space has dimension $D_n = 1 + n + C(n, 2)$. It is natural to ask whether adaptive dependency matching can produce a universal image-containment excess beyond D_n for other n , whether the excess is bounded or structured, or whether $n=6$ is exceptional. No generalization is claimed here.

19. Reproducibility

Repository: <https://github.com/RicardoVera-Lab/h6-universal-quadratic-image-theorem>

Minimal reproduction from the repository root:

```
python certificate/verify_h6_lower25_cleanroom.py certificate/h6_exact_profile_witnesses.txt
```

```
python certificate/verify_h6_publication_independent.py
```

```
certificate/h6_exact_profile_witnesses.txt
```

The proof-authoritative primary checker uses only exact integer/rational operations and the Python standard library. A successful rerun is evidence that the published finite certificate verifies as claimed; it should not be confused with independent proof review.

20. Acknowledgment of computational assistance

Computational and AI-assisted research tools were used during exploration, adversarial testing, code development, and manuscript preparation. No language-model output, stochastic sample, or opaque solver verdict is treated as mathematical proof authority. The claims in this paper are supported by the explicit human-readable reductions and exact certificate checks described above. Responsibility for the mathematical claims and the final manuscript remains with the author.

21. Conclusion

We have established, within a reproducible computer-assisted proof framework, that every set of at most 24 points in F_2^{12} is contained in the image of some coordinatewise quadratic map $F_2^6 \rightarrow F_2^{12}$. Thus $h(6) \geq 25$. The central mechanism is adaptive dependency matching: two unavoidable evaluation relations are selected so that they lie inside the much larger target dependency space. The finite certificate architecture reduces the computational burden to exact, independently checkable statements about 2-dimensional profiles, Delsarte bounds, and MacWilliams feasibility.

The next scientific step is not to strengthen the numerical bound prematurely, but to obtain competent external mathematical reproduction of the proof chain and then submit a hardened manuscript for peer review.

Appendix A. Exact Delsarte lower bounds

The exact certified lower bounds for the number of dangerous words in a full-support even $[N, 11, \geq 4]$ code are:

16 $\rightarrow$ 1011

17 $\rightarrow$ 938

18 $\rightarrow$ 910

19 $\rightarrow$ 9532/11

20 $\rightarrow$ 746

21 $\rightarrow$ 4033/5

22 $\rightarrow$ 48555/56

23 $\rightarrow$ 177445/212

24 $\rightarrow$ 1680251/1865

Each value is strictly greater than 640.

Appendix B. Dependency ledger

Human proof: definition handling; ϕ_2 factorization; kernel criterion; reduction to 24; translation normalization; target-kernel code; 11-dimensional even subcode; Hamming bounds; dangerous/safe split; sum-free bridge; Delsarte inequality derivation; coset separation; MacWilliams interpretation; all-one complement pairing.

Finite exhaustion: enumeration of the 46 dangerous profiles; integer MacWilliams feasibility closure.

Certificate checks: 45 exact-kernel witnesses; nine exact rational Delsarte certificates; exceptional all-one closure; adversarial tamper tests.
External theorem: large sum-free structure theorem in the binary case.

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